

Gambling-Related Psychological Predictors and Moderators of Attentional Bias Among Electronic Gaming Machine Players

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Objective: Attentional biases (ABs) have been shown to develop in the context of substance use disorders. Relatively less focus has been paid toward the development of ABs in behavioral addictions such as gambling disorder (GD). Furthermore, the psychological predictors and moderators of AB in GD remain unknown. The present study addressed these empirical gaps. **Methods:** Fifty-two non-GD electronic gaming machine (EGM) players, 25 GD-EGM players, and 61 non-gamblers completed measures of gambling-related behaviors and cognitions (problem gambling severity, cravings, expectancies, motives) and substance use and mental health (alcohol use severity, depression symptoms). The relationships between these constructs and AB for EGM images were then assessed using a free-viewing eye-tracking paradigm. **Results:** Non-GD EGM players and GD-EGM players attended to EGM images significantly more than neutral images (with the largest AB for the EGM players with GD). For all EGM players, gambling expectancies regarding the negative emotional impact of gambling and alcohol use severity were associated with greater AB. For non-GD EGM players only, AB was moderated by the anticipation aspect of gambling craving and the self-enhancement aspect of gambling expectancies. **Conclusion:** The results provide further evidence that ABs develop in the context of excessive gambling and are associated with gambling and psychological variables. The findings support the incentive-salience theory of ABs in gambling and provide a rationale for the development of AB modification programs in the treatment of gambling disorder. Given the predominantly white sample, our results may not generalize to individuals of other ethnicities.

Public Health Significance Statement

This study shows that individuals with gambling disorder are more likely to attend to gambling-related cues than individuals without gambling disorder. Positive expectations associated with gambling and greater alcohol use are related to increases in the time spent attending to gambling-related images.

Keywords: attentional bias, eye gaze tracking, gambling disorder, electronic gaming machines, motives

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An attentional bias (AB) is defined as preferential attention and dedication of cognitive resources to personally relevant stimuli (Crombez et al., 2013). In the context of addictive behaviors, an AB refers to heightened attention to addiction-related stimuli (e.g., casino imagery; a deck of cards) relative to neutral non-addictive stimuli (e.g., wildlife imagery; a stack of books). There is a growing body of empirical literature indicating the importance of ABs in the development and maintenance of addictions, including alcohol (Sharma et al., 2001), tobacco (Mogg & Bradley, 2002), and cocaine (Copersino et al., 2004). ABs for addictive substances have been associated with some of the hallmark characteristics of addiction such as cravings and poorer treatment outcomes (Field, 2005; Field et al., 2009; Marissen et al., 2006). According to incentive-sensitization theory (Robinson & Berridge, 2008), stimuli associated with addictive behavior take on an incentive salience and become “wanted” through conditioning processes. Over time, this results in heightened attention to addiction-related stimuli, which then

influences approach behaviors such as gambling. One implication of incentive-sensitization theory is that individuals who engage in addictions and do so problematically should exhibit a greater AB toward addiction-related cues. While the incentive-sensitization theory has been used to explain the development of substance use disorders (Berridge & Robinson, 2016), its key tenets have been less frequently examined in gambling disorder (GD) (Thomsen et al., 2014).

GD is defined as the persistent engagement in gambling despite significant negative consequences (American Psychiatric Association, 2013). In the most recent edition of the Diagnostic and Statistical Manual of Mental Disorders (American Psychiatric Association, 2013), GD is now included in the Substance-Related and Addictive Disorders section alongside traditional chemical addictions (e.g., alcohol, cannabis, opioids, stimulants). This reclassification reflects decades of empirical evidence indicating that gambling shares significant overlap in neurological, clinical, and behavioral characteristics with other substance use disorders (Mann et al., 2016). Given the similarities between gambling and substance use, it is perhaps unsurprising that ABs for gambling-related stimuli have been identified in those with GD (see Hønsi et al., 2013).

Hønsi et al. (2013) identified several common limitations of studies that examined ABs in GD. First, only two of the 11 studies in their review used direct measures of attention (e.g., eye-gaze tracking), with the remainder using indirect measures that rely on manual response latencies to infer the allocation of attention, such as the Stroop task and the dot-probe task (Field & Cox, 2008). A second limitation was the small sample sizes in many of these studies. The majority included less than 20 participants per group, and no study had a sample size greater than 40. Third, only five studies measured potential predictors of AB, such as craving, despite its association with attentional processes among substance users (Field et al., 2009).

Since the Hønsi et al. (2013), at least seven additional studies have examined ABs in gambling (Ciccarelli et al., 2016a, 2016b, 2019; Hudson et al., 2017), with three of them using direct measures of attention (Grant & Bowling, 2015; McGrath et al., 2018, 2020). Several of these studies also examined correlates associated with gambling-related ABs, including frequency of gambling and gambling attitudes and beliefs (Grant & Bowling, 2015), craving, emotion dysregulation, depression, anxiety, and stress (Ciccarelli et al., 2016a, 2019), and severity of alcohol use (Ciccarelli et al., 2019), with mixed results.

Taken together, the research to date suggests that ABs often develop in the context of GD. Yet few studies have investigated key psychological correlates of ABs using eye-gaze tracking, and for those studies, the results were mixed. Moreover, other important but unexamined psychological predictors of AB may exist in GD. Given the similarities between GD and substance use disorders, psychological characteristics associated with AB in substance use disorders may also be important in GD. For example, evidence from the alcohol literature has revealed a strong relationship between positive outcome expectancies and AB, as measured by eye-gaze tracking (e.g., Field et al., 2011; Jones et al., 2012). As such, positive expectancies associated with gambling may also be an important predictor of AB in GD.

The Present Study

The purpose of the present study was to further current understanding of gambling-related ABs by examining how EGM players with GD and without GD differ from each other, and from non-gamblers, in their attention to EGM-related stimuli. Individuals whose gambling was primarily on EGMs were chosen for two reasons: EGM gambling is most commonly linked to GD (Delfabbro et al., 2020), and previous research by our team highlights the importance and advantage of matching gambling stimuli with preferred gambling activities in eye-tracking research (McGrath et al., 2018). ABs were assessed using a free-viewing eye-gaze tracking paradigm, and the stimuli were a mixture of EGM images and neutral (non-gambling) images. The psychological correlates were chosen based on previous research suggesting their association with AB and substances (i.e., craving, expectancies, alcohol use severity, symptoms of depression) and for exploratory purposes (e.g., gambling motives).

We also explored some of these correlates as potential moderators. Gambling-related craving was assessed given the relationship between craving and attentional bias in substance use disorders (Field et al., 2009) and, to some extent, in GD (Ciccarelli et al., 2016a). We assessed expectancies for gambling because individuals who hold a positive outlook for gambling may be more likely to preferentially attend to gambling-related images. Alcohol use and depressive symptoms were also assessed given they commonly co-occur with GD (Dowling et al., 2015; Lorains et al., 2011) and have been associated with increased severity of GD (Martens et al., 2009; Schluter et al., 2019).

Based on the extant literature, we hypothesized that compared to non-gamblers, EGM players would exhibit a greater AB toward EGM images. In addition, we hypothesized that GD-EGM players would have the largest AB for EGM images. No specific a priori hypotheses were made regarding the psychological predictors and moderators of AB, given the relative paucity of previous research and mixed findings. Incentive sensitization theory would suggest that given the incentive salience (“liking”) attributed to gambling with repeated engagement, AB could be associated with positive reinforcement expectancies of gambling (such as enjoyment, self-enhancement, anticipation, and desire). It is also possible that with significant over-involvement in gambling, the hedonistic component of “liking” is diminished, whereas the compulsive nature of “wanting” remains, and thus the positive reinforcement expectancies may not be present in those with GD. One of the advantages of our methodology is the ability to distinguish between these two possibilities.

Method

Participants

One hundred thirty-seven participants (52 non-GD EGM players, 25 GD-EGM, and 60 non-gamblers) were recruited from the local community ($n = 87$) through online advertisements (e.g., Facebook ads, Kijiji) and through an institutional research participant system ($n = 50$). Institutional ethics approval was obtained prior to data collection. Prospective participants first completed a phone screen to determine their eligibility. The inclusion criteria for EGM players were as follows: (a) adults (18+ years old) and (b) regular EGM players (e.g., slot machines, video lottery terminals), defined as

having played an EGM for money at least once per month for the past 3 months (McGrath et al., 2012). All participants were asked to choose their preferred type of gambling from a list of options. Non-gamblers indicated that they had never gambled on an EGM in their lifetime and that they had not gambled for money in the past 12 months.

Exclusion criteria for all participants were as follows: (a) currently seeking treatment for gambling-related problems or had done so in the past, (b) intending to quit gambling in the next 30 days, and (c) color blindness. Additional exclusion criteria for the eye-tracking data collection were: (a) inability to complete the calibration procedure and (b) inability to achieve adequate pupil and corneal reflection (resulting in poor quality data).

Stimuli

The experimental stimuli were 336 high-resolution images that were displayed across 84 test trials and two practice trials. Images were obtained from stock-photo subscription websites. There were two categories of images: EGM-related images and neutral images. The EGM images were selected to reflect a range of EGM terminals and focused on the interfaces of individual machines and rows of machines. Care was taken to avoid the inclusion of people in the images (with the exception of hands interacting with an EGM) and substances such as alcohol. Neutral images included pictures of arcade games (such as pinball machines), terminals used in manufacturing plants, and data storage systems such as large hard drive towers. The neutral images were chosen to match the EGM images on structural characteristics, color, and brightness. Some images were modified using editing software to better match EGM stimuli (e.g., by altering the hue of an image).

On each trial, four images were presented on a neutral background, with each image measuring 300 pixels in height and 400 pixels in width. Of the 84 trials, 28 consisted of three neutral images and one EGM image; these were the experimental trials. For the remaining 56 trials (filler trials), the displays consisted of four neutral images only (no EGM images). The filler trials were not used in the analyses. The filler trials served to decrease the number of EGM images displayed during the eye-tracking task in order to reduce potential demand characteristics (e.g., participants hypothesizing that they should focus on EGM images).

Measures

Demographics

Participant demographic information was assessed using a questionnaire developed from the Statistics Canada census forms. Data collected included participants' age, sex, education level, employment status, annual household income, marital status, and ethnicity.

Gambling

Gambling behavior was measured with a 90-day Gambling Timeline Follow-Back Questionnaire (G-TLFB; Weinstock et al., 2004). The TLFB is a calendar-based interview designed to obtain information on various aspects of an individual's gambling behavior (e.g., time spent gambling, money spent gambling, etc.). Participants are asked to recall their gambling episodes by viewing a

calendar that includes personalized events (e.g., holidays, work schedules, etc.). Events serve as a memory cue to aid in the recall of the days they gambled and the details of their gambling episodes. The G-TLFB has adequate to excellent test-retest reliability and strong concurrent and discriminant validity (Weinstock et al., 2004). The number of days and the time spent playing EGMs was positively correlated, $r(83) = .87, p < .001$. The number of days playing EGMs was negatively correlated with net wins/losses, $r(82) = -.57, p < .001$, and with time spent playing EGMs, $r(82) = -.76, p < .001$.

Problem Gambling Severity

The severity of GD was assessed with the Problem Gambling Severity Index (PGSI) from the Canadian Problem Gambling Index (CPGI; Ferris & Wynne, 2001). The PGSI contains nine items (e.g., "Have you felt that you might have a problem with gambling?") that are rated from 0 (*never*) to 3 (*almost always*). Individual items are summed to obtain an overall score, with higher scores indicating a higher likelihood of GD. Based on the analyses and recommendations of Currie et al. (2013), participants with scores from 0 to 4 were classified as non-GD, and participants with scores of 5 or more were classified as GD. The PGSI is one of the most widely used measures of problem gambling severity and has strong psychometric properties in assessing problem gambling symptoms, including test-retest reliability in the interpretative categories (Currie et al., 2013). For the current sample, the internal consistency (Cronbach's alpha, a measure of scale reliability) of the PGSI was very good ($\alpha = .89$).

Gambling Craving

Gambling-related cravings were assessed using the Gambling Craving Scale (GACS; Young & Wohl, 2009). The GACS is a multidimensional measure of gambling craving. For each of the nine statements, participants indicate the extent to which they agree or disagree, using a scale from 0 (*strongly disagree*) to 7 (*strongly agree*). The GACS is comprised of three subscales: Desire, Anticipation, and Relief. The Desire subscale assesses a strong and urgent desire to gamble (e.g., "I crave gambling right now"). Anticipation represents an intention to gamble that is anticipated to be fun and enjoyable (e.g., "Gambling would be fun right now"), and the Relief subscale assesses relief from negative experiences expected from gambling (e.g., "Gambling would make me less depressed"). The three-factor structure has been reliably demonstrated in various samples, and the GACS demonstrates concurrent validity with gambling-related measures (Young & Wohl, 2009). For the current sample, the internal consistency of the GACS was excellent ($\alpha = .91$) and similarly strong for the Desire ($\alpha = .90$), Anticipation ($\alpha = .84$), and Relief ($\alpha = .86$) subscales.

Gambling Expectancies

The Gambling Expectancy Questionnaire (GEQ; Gillespie et al., 2007) is a 23-item instrument designed to measure the extent to which the expectation of positive or negative outcomes related to gambling influence gambling behaviors (e.g., "I will not be able to stop"). Each item is rated on a 7-point Likert scale ranging from 1 (*no chance*) to 7 (*certain to happen*). The GEQ contains five

subscales, each capturing a positive or negative expectancy associated with gambling, with three positive expectancies (enjoyment/arousal, self-enhancement, money) and two negative expectancies (over-involvement, emotional impact). The five factors have demonstrated adequate to good internal reliability (Gillespie et al., 2007). For the current sample, the internal consistency of the total scale was excellent ($\alpha = .92$), although the subscales varied in their reliability (enjoyment/arousal = .95, self-enhancement = .82, money = .62, over-involvement = .68, emotional impact = .79).

Gambling Motives

The Gambling Motives Questionnaire-Financial (GMQ-F; Dechant, 2014), which was derived from the original Gambling Motives Questionnaire (Stewart & Zack, 2008), was used to assess participants' motivations for gambling. The GMQ-F consists of four subscales that correspond to coping motives (e.g., "To forget your worries"), financial motives (e.g., "To win money"), social motives (e.g., "To be sociable"), and enhancement motives (e.g., "Because it's exciting"). The items are anchored from 1 (*never/almost never*) to 4 (*always/almost always*), with higher scores indicating a greater degree of gambling for the specific motive. The four-factor structure of the GMQ-F has been documented in different samples (Schellenberg et al., 2016). For the current sample, the internal consistency for the total scale was good ($\alpha = .80$), whereas the four subscales varied in their reliability (coping motives = .85, financial motives = .74, social motives = .68, and enhancement motives = .84).

Alcohol Use Severity

The Alcohol Use Disorders Identification Test (AUDIT; Saunders et al., 1993) was used to measure participants' alcohol use severity. The AUDIT consists of 10 items and assesses both the frequency of alcohol consumption (e.g., "How often do you have six or more drinks on one occasion?") and the consequences of alcohol use (e.g., "How often during the last year have you had a feeling of guilt or remorse after drinking?"). Items are anchored from 0 to 4, with higher scores indicating greater alcohol use severity. The AUDIT is one of the most widely used measures of alcohol use severity and has excellent psychometric properties compared to other self-report measures of alcohol severity (Reinert & Allen, 2007). For the current sample, the internal consistency of the AUDIT was good ($\alpha = .86$).

Depressive Symptoms

The Beck Depression Inventory-Second Edition (BDI-II; Beck et al., 1996) was used to assess depression symptom severity. The BDI-II consists of 21 items that measure symptoms of depression. For each item, the participant chooses the statement that best describes their status during the past 2 weeks (e.g., "I am so sad and unhappy, I can't stand it"). Each item is scored on a scale from 0 to 3, with total scores ranging from 0 to 63, which reflect minimal depression (0–13), mild depression (14–19), moderate depression (20–28), and severe depression (29–63). The BDI-II has excellent internal consistency and test-retest reliability. For the current sample, the internal consistency of the BDI-II was excellent ($\alpha = .90$).

Eye-Tracking Measures

Two eye-tracking measures were used to assess participants' attention to the images: time of first fixation and total fixation time. The time of first fixation was measured as the amount of time that elapsed (in ms) from the beginning of the 6-s presentation until an image was fixated, calculated separately for EGM images and neutral images on a trial-by-trial basis, and then averaged over the 28 experimental trials. The time of first fixation is a measure of participants' initial attention to the images and was used to assess participants' relative interest in the images at the beginning of the presentation. Each of the experimental trials showed one EGM image and three neutral images, so the time of first fixation for the neutral images was calculated as the average time of first fixation for the three neutral images. The second eye-tracking measure (total fixation time) was the total amount of time (in ms) spent fixating on an image during the 6-s presentation. Total fixation time was calculated separately for EGM and neutral images on a trial-by-trial basis and then averaged over the 28 experimental trials. Total fixation time for the neutral images was calculated as the average total fixation time for the three neutral images. Longer total fixation times reflect greater attention to an image over the 6-s presentation and are, therefore, a measure of sustained attention. Total fixation times to valenced and neutral images collected in a free-viewing eye-tracking paradigm have been shown to have strong internal consistency and good test-retest reliability (Sears et al., 2019).

We created a measure of AB for each participant by calculating the difference (in ms) between the mean total fixation time for EGM images and the mean total fixation time for neutral images. A positive bias (significantly greater than 0 ms) indicated that a participant attended to EGM images more than neutral images, whereas a negative bias (significantly less than 0 ms) indicated that a participant attended to neutral images more than EGM images. A bias not significantly different than 0 ms indicated that a participant attended to the EGM and neutral images equally. *t*-Tests were used to determine if the mean bias for each group differed significantly from 0 ms.

Eye-Tracking Apparatus

Eye movements were recorded using an EyeLink 1,000 eye-tracking system (SR Research Ltd, Ottawa, Ontario, Canada), which uses infrared video-based tracking technology. The system has a 1,000-Hz sampling rate, a temporal resolution of 1 ms, and an average gaze error of less than 0.5° of visual angle. Stimuli were shown on a 24-in. LCD monitor positioned approximately 60 cm away from the participant. To increase tracking accuracy, participants used a chin rest to keep viewing distance constant and to minimize head movements while they viewed the images. Prior to data collection, a calibration procedure was completed to ensure accurate gaze measurement.

Procedure

At the beginning of their lab visit, participants received information about the study procedures and provided written informed consent. Participants viewed a total of 84 trials (including two practice trials), each trial consisting of four images presented for 6 s. The position of each image was fixed to one of the four quadrants

of the display. Participants were informed that they would have 6 s to view the images and that they could view the images in any way they desired (i.e., a free-viewing paradigm). Eye-gaze tracking data was collected throughout each 6-s presentation. Twenty-eight trials consisted of one EGM image and three neutral images (the data trials), and 56 trials consisted of four non-gambling images (the filler trials). For the trials with an EGM image, the EGM image was presented in each quadrant an equal number of times (i.e., seven times in the upper left, seven times in the upper right, seven times in the lower left, seven times in the lower right). The filler trials were used to reduce the relative frequency of EGM images during the task (i.e., the majority of the images were neutral images), thereby increasing their novelty and salience (the eye-tracking data from the filler trials were not analyzed). The order of the trials was randomized across participants.

After viewing the images (which took approximately 10 min), participants completed the measures of interest (described previously) and were fully debriefed. Participants were provided with a \$40 (CAD) gift card as compensation.

Processing of Eye-Tracking Data

The fixation data were processed using the EyeLink Data Viewer analysis software (SR Research) to filter for blinks, missing data, and other recording artifacts (using the default settings). To be included in the analyses, a fixation had to be at least 100 ms in duration; sequential, adjacent fixations that were less than 100 ms were merged into a single fixation.

Eye-tracking data were collected from 149 individuals. The data from 12 participants (8.05% of the data collected) were excluded due to data quality issues. More specifically, one participant was excluded due to difficulties with calibration, four were excluded due to poor tracking, one was excluded because of unusually long total fixation times, and six were excluded because they reported difficulty maintaining their alertness and concentration during data collection. As such, the final sample consisted of 137 participants.

Analytic Plan

All statistical analyses were carried out using the Statistical Package for the Social Sciences (SPSS, version 26, IBM Corp, 2019). Analysis of variance (ANOVA) and chi-square tests were used to compare the demographic characteristics of non-gamblers, non-GD EGM players, and GD EGM players. Factorial analysis of variance and *t*-tests were used to determine if groups differed in their time of the first fixation to EGM and neutral images. The AB variable was analyzed using one-factor ANOVA and *t*-tests. For all *t*-tests we used an alpha of 1%. Effect sizes are reported for all statistical comparisons (Cohen's *d* for *t*-tests and partial η^2 for *F* tests). For Cohen's *d*, a value of 0.2 is conventionally interpreted as representing a small effect, 0.5 a moderate effect, and 0.8 a large effect; for partial η^2 , values of 0.01, 0.06, and 0.14 are conventionally interpreted as small, medium, and large effects, respectively (Cohen, 1969, 1992). Statistical power analyses for the ANOVA analyses were carried out using G*Power 3.1 (Faul et al., 2007). Based on recent eye-tracking studies of AB in similar samples of gamblers (McGrath et al., 2018, 2020), we expected a fairly large effect size in the between-group comparison of AB (e.g., using the same free-viewing paradigm, McGrath et al., 2020, reported a partial η^2 of 0.26 in a comparison of their no-risk, low-risk, and moderate/high-risk groups). To be conservative, we assumed a partial η^2 of 0.14 in our power

calculations and these showed that with an alpha of 5%, power would be 87% with 25 participants per group (our final sample size exceeded this criterion, as there were 61 non-gamblers, 52 EGM players without GD, and 25 EGM players with GD in the ANOVA analyses).

Bivariate correlations (Pearson's *r*) were used to examine correlates of AB and the measures of interest (gambling craving, gambling expectancies, gambling motives, alcohol use severity, depressive symptoms) among the 75 EGM players. Hierarchical linear regression analyses were used to determine if the ABs of GD and non-GD EGM players were moderated by gambling-focused cognitions (e.g., cravings, expectancies, as measured by the GACS and GEQ). A power analysis was conducted to assess the power of the moderation analyses. For the sample of 137 participants, the power to detect a 5% increase in the variance accounted for (i.e., increase in R-square) in the outcome variable by the interaction effect was 76% (a small effect size; $f^2 = 0.05$; for a medium effect size, $f^2 = 0.15$, power was 99%). Given the exploratory nature of the correlational and moderation analyses we did not control for Type I error inflation and this should be kept in mind when interpreting our results.

Results

Demographics

The sample consisted of 70 women and 67 men, with a mean age of 31.02 years ($SD = 12.14$). The majority of participants reported their ethnicity as White ($n = 97$, 71.3%). Eleven participants identified as Aboriginal, seven identified as South Asian, seven as Latin American, four identified as being Black, four as Arab, two as Chinese, two as "Other", and one each as Filipino, South East Asian, West Asian, Japanese, and Korean. One participant did not report ethnicity. With respect to employment status and education, 24.8% of the sample reported being employed full-time, 36.5% reported being a student, and 98.5% reported their highest level of education as high school or greater.

Of the 77 EGM players, 52 (67.53%) were classified as non-GD (i.e., a score of 0–4 on the PGSI) and 25 (32.46%) were classified as GD-EGM players (a score of 5 or greater on the PGSI). The non-GD EGM players reported they played EGM machines an average of 2.63 days a month ($SD = 2.39$), and the GD EGM players played an average of 4.46 days a month ($SD = 3.84$). The non-GD EGM players reported spending an average of \$221 (CAD) per month ($SD = 689.98$) on EGM machines, whereas for GD-EGM players, the average was \$427 ($SD = 441.84$).

With respect to demographic differences (Table 1), GD-EGM players were significantly older than non-GD EGM players and non-gamblers. GD-EGM players were also less likely to be employed full-time or to be a student and reported lower income than non-GD EGM players and non-gamblers. There were no differences between the groups for age, gender, ethnicity, or education. As expected, the groups differed significantly on the gambling and psychological variables (Table 2). GD-EGM players reported significantly higher gambling expectancies (over-involvement, self-enhancement, emotional impact), gambling motives (coping, financial), and gambling craving (anticipation, desire). Non-GD EGM players reported the highest level of problematic alcohol use, although no differences were found in post-hoc analyses. The groups did not differ in their levels of depression symptoms.

Table 1*Demographic Characteristics and Comparisons Between Non-EGM Players, Non-GD EGM Players, and GD-EGM Players*

Characteristic	Non-gamblers, <i>n</i> = 60	Non-GD, <i>n</i> = 52	GD, <i>n</i> = 25	χ^2	<i>F</i>	<i>p</i>	Effect size
Age in years, mean (<i>SD</i>)	26.9 (8.7)	30.7 (12.1)	41.7 (13.2)		15.58	<.001	.191 ^a
Gender, <i>n</i> (%)				.241		.886	.042 ^b
Male	28 (46.7)	26 (50.0)	13 (52.0)				
Female	32 (53.3)	26 (50.0)	12 (48.0)				
Ethnicity, <i>n</i> (%)				1.67		.431	.110 ^b
White	41 (68.3)	40 (78.4)	16 (64.0)				
Non-white	19 (31.7)	12 (21.6)	9 (36.0)				
Marital status, <i>n</i> (%)				4.25		.118	.150 ^b
Married/common-law	9 (15.0)	15 (28.9)	8 (32.0)				
Not married/common-law	51 (85.0)	37 (71.1)	17 (68.0)				
Highest level of education, <i>n</i> (%)				2.17		.336	.126 ^b
High school	34 (56.6)	29 (55.7)	10 (40.0)				
More than high school	26 (43.4)	23 (44.3)	15 (60.0)				
Full-time employment, <i>n</i> (%)				10.37		.001	.275 ^b
Yes	7 (11.7)	17 (32.7)	10 (40.0)				
No	53 (88.3)	35 (67.3)	15 (60.0)				
Student, <i>n</i> (%)				15.46		<.001	.336 ^b
Yes	35 (58.3)	22 (42.3)	3 (12.0)				
No	25 (41.7)	30 (57.7)	22 (88.0)				
Household income, <i>n</i> (%)				17.80		.007	.257 ^b
\$0–\$25,000	19 (32.2)	4 (7.8)	8 (32.0)				
\$25,000–\$50,000	10 (16.9)	11 (21.6)	9 (36.0)				
\$50,000–\$100,000	17 (28.8)	20 (39.2)	7 (28.0)				
More than \$100,000	13 (22.0)	16 (31.4)	1 (4.0)				

Note. EGM = electronic gaming machine; Non-GD = nongambling disorder electronic gaming players; GD = gambling disorder electronic gaming players.

^a Partial η^2 .

^b Cramer's *V*.

Analysis of Time of First Fixation

Figure 1 shows the mean time of first fixation (in ms) for EGM images and neutral images, for non-gamblers, non-GD EGM players, and GD-EGM players. These data were analyzed using a mixed-model

ANOVA, with the factors Group (non-gamblers, non-GD EGM players, GD-EGM players) and Image Type (EGM image, neutral image). The main effect of Group was not significant, $F(2, 134) = 0.35$, $p = .69$, partial $\eta^2 = .005$. There was a main effect of

Table 2*Comparisons of Gambling Measures, Alcohol Use Severity, and Depression Symptoms by EGM Gambling Status*

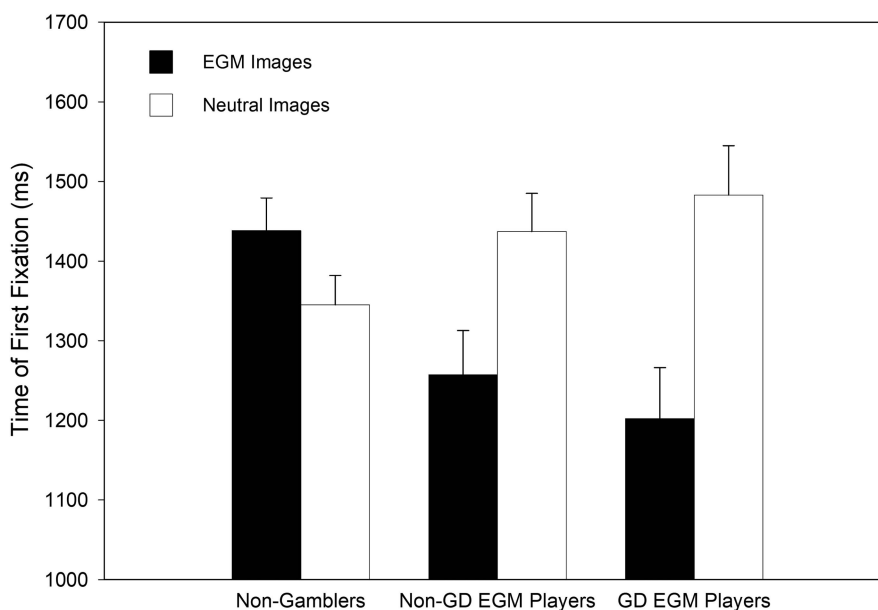
Measure	Non-gamblers (<i>n</i> = 60) <i>M</i> (<i>SD</i>)	Non-GD (<i>n</i> = 52) <i>M</i> (<i>SD</i>)	GD (<i>n</i> = 25) <i>M</i> (<i>SD</i>)	<i>F</i>	<i>p</i>	Effect size
GEQ						
Enjoyment/arousal	2.64 (1.23)	5.13 (0.58)	5.13 (0.80)	109.04	<.001	.636
Overinvolvement	2.49 (1.17)	2.64 (1.08)	3.40 (0.98)*	5.89	.004	.086
Self-enhancement	1.24 (0.71)	1.97 (0.83)	2.75 (1.12)*	27.33	<.001	.304
Emotional impact	2.20 (1.25)	3.12 (1.09)	3.69 (0.93)*	16.54	<.001	.209
Money	1.67 (0.95)	2.70 (0.98)	2.98 (1.33)	18.12	<.001	.225
GMQ-F						
Coping	—	1.34 (0.49)	1.80 (0.73)	8.93	.004	.123
Financial	—	2.28 (0.81)	2.78 (0.70)	6.09	.016	.087
Social	—	2.08 (0.54)	1.91 (0.61)	1.41	.239	.022
Enhancement	—	2.44 (1.10)	2.70 (0.65)	1.10	.298	.017
GACS						
Anticipation	1.87 (1.12)	4.28 (1.38)	5.06 (1.25)*	79.33	<.001	.542
Desire	1.04 (0.27)	2.01 (1.33)	2.81 (1.47)*	28.20	<.001	.296
Relief	1.08 (0.37)	1.89 (1.29)	2.40 (1.52)	16.06	<.001	.193
AUDIT	3.68 (4.20)	8.78 (6.09)	8.16 (7.47)	12.78	<.001	.160
BDI	9.45 (8.51)	10.75 (8.82)	13.00 (8.70)	1.49	.228	.022

Note. EGM = electronic gaming machine; Non-GD = nongambling disorder electronic gaming players; GD = gambling disorder electronic gaming players; Effect size = partial η^2 from ANOVA; GEQ = gambling expectancy questionnaire; GMQ-F = gambling motives questionnaire-financial; GACS = gambling craving scale; AUDIT = alcohol use disorders identification test; BDI = Beck depression inventory. The sample size for the GMQ-F was 43 non-GD and 23 GD.

* Statistically significant difference between the non-GD EGM players and the GD EGM players according to a *t*-test ($p < .05$).

Figure 1

Mean Time of First Fixation (in ms) for EGM Images and Neutral Images, for Non-Gamblers, Non-GD EGM Players, and GD EGM Players. Error Bars Show One Standard Error of the Mean



Note. EGM = electronic gaming machines; GD = gambling disorder.

Image Type, $F(1, 134) = 31.56, p < .001$, partial $\eta^2 = .19$, which was qualified by a Group $\times$ Image Type interaction, $F(2, 134) = 29.50, p < .001$, partial $\eta^2 = .31$.

As can be seen in Figure 1, the nature of the interaction suggests that non-gamblers fixated on neutral images earlier in the presentations than EGM images, whereas EGM players fixated on EGM images earlier than neutral images. Within-subjects t -tests confirmed these hypothesized differences. More specifically, for non-gamblers, first fixation times for neutral images (1,345 ms) were significantly shorter than first fixation times for EGM images (1,438 ms), $t(59) = 3.29, p = .002, d = 0.42$. In contrast, for non-GD EGM players, first fixation times for EGM images (1,257 ms) were significantly shorter than first fixation times for neutral images (1,437 ms), $t(51) = 5.01, p < .001, d = 0.69$, which was also the case for GD-EGM players (1,202 ms vs. 1,483 ms), $t(24) = 5.98, p < .001, d = 1.20$ (medium and large effect sizes, respectively).

Additional between-group t -tests showed that non-GD and GD-EGM players' first fixations to EGM images were significantly shorter than the first fixations of non-gamblers, $t(110) = 2.63, p = .011, d = 0.49$; $t(83) = 3.07, p = .003, d = 0.73$, respectively. The non-GD and GD EGM players' first fixation times did not differ, $t(75) = 0.59, p = .554, d = 0.15$. Together these results indicate that EGM players attended to the EGM images in the displays before they attended to the neutral images, unlike the non-gamblers.

Attentional Bias

As predicted, the GD-EGM players had a large AB for EGM stimuli ($M = 855.38$ ms, $SD = 522.92$). That is, GD-EGM players attended to EGM images more than neutral images (855.38 ms longer, on average) and this difference was significantly greater than

zero, $t(24) = 8.17, p < .001, d = 1.63$. The AB of the non-GD EGM players was smaller ($M = 381.46$ ms, $SD = 550.31$) but also significantly greater than zero, $t(51) = 4.99, p < .001, d = 0.69$, although the effect size was much smaller. In contrast, the AB of the non-gamblers was significantly less than zero ($M = -231.18$ ms, $SD = 297.73$), $t(59) = 6.01, p < .001, d = 0.78$, which indicated that non-gamblers attended to neutral images significantly more than EGM images.

The ABs of the three groups were compared using a one-factor between-groups ANOVA, which revealed a significant effect of Group, $F(3, 134) = 58.065, p < .001$, partial $\eta^2 = .46$. Follow-up between-group t -tests showed that, as predicted, the AB of GD-EGM players was significantly larger than the AB of non-gamblers, $t(83) = 10.12, p < .001, d = 2.55$, and the AB of non-GD EGM players was significantly larger than the AB of non-gamblers, $t(110) = 7.17, p < .001, d = 1.38$. In addition, the GD-EGM players had a significantly larger AB than the non-GD EGM players (855.38 ms vs. 381.46 ms), $t(75) = 4.31, p < .001, d = 0.88$ (a large effect size), which indicated that greater problem gambling severity was associated with a larger AB.

Psychological Correlates of Attentional Bias

Table 3 shows the intercorrelations between the variables of interest, and Table 4 shows the correlations between AB and the key measures. For the full sample, statistically significant correlations were observed for all the variables of interest, with the exception of the GMQ-F social motives subscale. For non-GD EGM players, AB had significant positive correlations with the gambling expectancies of enjoyment/arousal, self-enhancement,

Table 3*Intercorrelations Between the Measures for the Full Sample (N = 137)*

Measure	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. GEQ-enjoyment	—	.27*	.40**	.52**	.52**	.25*	.30*	.32*	.35*	.71**	.40**	.35**	.43**	.11
2. GEQ-overinvolvement		—	.44**	.19*	.41**	.50**	.30*	-.11	.19	.26*	.28*	.21*	.18*	.23*
3. GEQ-self enhancement			—	.35**	.48**	.53**	.37*	-.10	.07	.51**	.52**	.48**	.31**	.26*
4. GEQ-emotional impact				—	.37**	.28*	.47**	.00	.12	.46**	.44**	.39**	.24*	.17
5. GEQ-money					—	.33*	.24*	-.03	.13	.49**	.34**	.32**	.27**	.16
6. GMQ-F coping						—	.40*	-.09	.23	.43**	.46**	.57**	.22	.39*
7. GMQ-F financial							—	.14	.18	.34*	.34*	.38*	.11	.34*
8. GMQ-F social								—	.09	-.05	-.17	-.15	.10	-.07
9. GMQ-F enhancement									—	.36*	.06	.22	-.09	-.05
10. GACS anticipation										—	.65**	.58**	.40**	.13
11. GACS desire											—	.72**	.33**	.24*
12. GACS relief												—	.21*	.25*
13. AUDIT													—	.27*
14. BDI														—

Note. GEQ = gambling expectancy questionnaire; GMQ-F = gambling motives questionnaire-financial; GACS = gambling craving scale; AUDIT = alcohol use disorders identification test; BDI = Beck depression inventory.

* $p < .05$. ** $p < .001$.

and emotional impact (GEQ subscales), the enhancement and financial aspects of gambling motives (GMQ-F subscales), with the anticipation, desire, and relief aspects of gambling craving (GAC subscales), and with depression symptoms (BDI scores) and alcohol use severity (AUDIT scores). In contrast, for the GD-EGM players AB was positively correlated only with GEQ emotional impact and alcohol use severity

Moderation Analyses

Hierarchical multiple regression was conducted to test for statistically significant moderation (interaction) effects using PROCESS

(version 3.4.1; Hayes, 2018; Hayes & Rockwood, 2017). For each analysis, AB was the dependent variable, and Group (non-EGM player, non-GD EGM player, GD EGM player) was the categorical predictor variable (using indicator coding). Mean-centered subscale scores on the GACS and GEQ, and mean-centered scores on the AUDIT and BDI, were used as the moderator variable in separate analyses. Summary results for all the moderation analyses are listed in Table 5. Note that we did not carry out analyses with the GMQ-Financial subscale scores because only the non-GD and GD EGM players completed this measure ($N = 77$); therefore, the statistical power to detect interactions was relatively low.

Table 4*Correlations (With 95% Confidence Intervals) Between AB and Gambling and Psychological Measures*

Measure	Total	Non-gamblers	Non-GD	GD
GEQ				
Enjoyment/arousal	.57 (.49, .65)***	.08 (-.17, .36)	.30 (.02, .56)*	.32 (-.07, .623)
Overinvolvement	.22 (.06, .39)**	-.11 (-.38, .16)	.20 (-.06, .45)	.01 (-.35, .34)
Self-enhancement	.47 (.32, .60)***	.13 (-.11, .35)	.39 (.13, .60)**	-.20 (-.63, .22)
Emotional impact	.46 (.32, .59)***	.08 (-.17, .39)	.31 (.05, .53)*	.34 (.05, .63)*
Money	.37 (.23, .50)***	.14 (-.09, .39)	.17 (-.07, .39)	-.05 (-.40, .29)
GMQ-F				
Coping	.33 (.08, .56)**	—	.19 (-.20, .524)	.27 (-.25, .65)
Financial	.32 (.11, .53)**	—	.33 (.04, .57)*	.00 (-.44, .44)
Social	.16 (-.09, .38)	—	.23 (-.08, .47)	.29 (-.10, .60)
Enhancement	.25 (.08, .51)*	—	.26 (.07, .62)*	.04 (-.42, .47)
GACS				
Anticipation	.62 (.51, .72)***	.25 (.02, .50)*	.43 (.19, .62)**	-.11 (-.44, .31)
Desire	.47 (.31, .62)***	.23 (.10, .38)*	.24 (-.04, .57)*	.00 (-.37, .32)
Relief	.45 (.28, .61)***	-.07 (-.25, .21)	.37 (.02, .73)***	.07 (-.26, .43)
AUDIT	.42 (.26, .59)***	.16 (-.18, .48)	.26 (.04, .49)*	.39 (-.08, .72)*
BDI	.22 (.05, .38)**	.06 (-.19, .32)	.31 (.02, .57)*	.03 (-.36, .38)

Note. AB = attentional bias; Non-GD = non-gambling disorder electronic gaming players; GD = gambling disorder electronic gaming players; GEQ = gambling expectancy questionnaire; GMQ-F = gambling motives questionnaire-financial; GACS = gambling craving scale; AUDIT = alcohol use disorders identification test; BDI = Beck depression inventory. The sample size for the GMQ-F was 43 non-GD and 23 GD. Confidence intervals were derived using bootstrapping with 1000 samples.

* $p < .05$, one-tailed significance. ** $p < .01$. *** $p < .001$, one-tailed significance.

Table 5

Summary of Hierarchical Linear Regression Analyses Testing for Interactions Between Group (Non-Gamblers, Non-GD EGM Players, GD EGM Players) and GEQ, GACS, AUDIT, and BDI Scores on the Prediction of Attentional Bias

Measure	<i>R</i>	<i>R</i> ²	ΔR^2	<i>F</i>	<i>p</i>
GEQ					
Enjoyment/arousal	.704	.496	.026	3.18	.044
Overinvolvement	.682	.465	.011	1.26	.285
Self-enhancement	.713	.509	.043	5.34	.005
Emotional impact	.707	.501	.019	2.34	.100
Money	.680	.463	.005	0.67	.510
GACS					
Anticipation	.726	.528	.027	3.76	.025
Desire	.698	.487	.009	1.22	.295
Relief	.712	.507	.016	2.15	.119
AUDIT	.713	.508	.003	0.41	.661
BDI	.703	.495	.014	1.88	.155

Note. Non-GD = nongambling disorder electronic gaming players; EGM = electronic gaming machine; GD = gambling disorder electronic gaming players; GEQ = gambling expectancy questionnaire; GMQ-F = gambling motives questionnaire-financial; GACS = gambling craving scale; AUDIT = alcohol use disorders identification test; BDI = beck depression inventory. *R* and *R*² are the statistics for the full model (two predictors and their interaction). ΔR^2 is the increase in *R*² due to the interaction.

As shown in Table 5, there were three statistically significant moderation effects identified, involving the GEQ enjoyment/arousal subscale, the GEQ self-enhancement subscale, and the GACS anticipation subscale. The nature of these interactions can be seen in Figures 2–4, which show the relationship between AB and the subscale scores for each of the three groups (at the mean subscale score, one standard deviation below the mean, and one standard deviation above the mean, as calculated by PROCESS, and labelled as “low,” “average,” and “high” in the figures).

As can be seen in Figure 2, for the GEQ enjoyment/arousal subscale analysis, the interaction reflected that higher GEQ enjoyment/arousal scores were associated with larger ABs, but only for non-GD and GD-EGM players. The positive association was quite similar for the two EGM player groups, which suggests that the interaction was due to the inclusion of the non-gamblers in the analysis. To determine if this was the case, the identical analysis was carried out with the data of only the non-GD and GD-EGM players. In this analysis, the interaction was not statistically significant, $F(1, 73) = 0.172$, $p = .679$, as it increased the percentage of variance accounted for by only 0.18%. Thus, the positive association between GEQ enjoyment/arousal scores and AB did not differ for non-GD and GD-EGM players and was therefore not considered further.

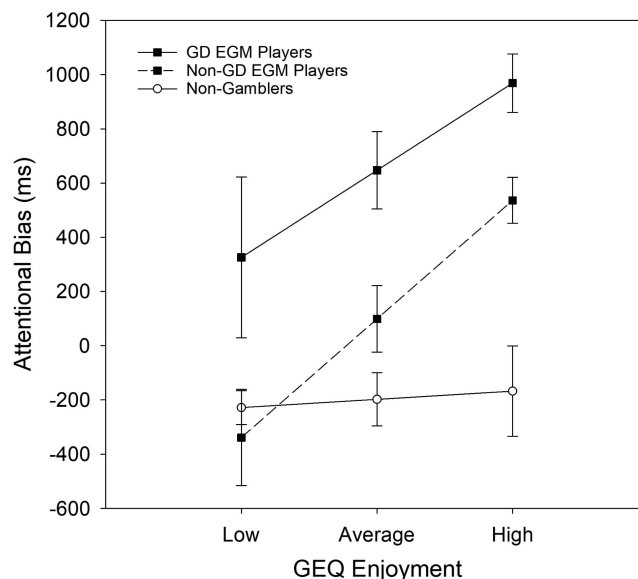
Figure 3 shows the relationship between GEQ self-enhancement subscale scores and AB for each of the three groups. As can be seen in Figure 3, for non-GD EGM players, higher self-enhancement scores were associated with a larger AB, whereas for GD-EGM players and non-gamblers, higher scores had a much smaller effect. An additional analysis of the data of only the non-GD and GD-EGM players confirmed that the interaction was present when the analysis was confined to EGM players, $F(1, 73) = 7.616$, $p = .007$ ($\Delta R^2 = 7.83\%$). A simple slopes analysis showed that for non-GD EGM players, higher self-enhancement scores were associated

with a larger AB, $t(51) = 3.00$, $p = .003$, $\beta = .392$, whereas for GD-EGM players this was not the case, $t(24) = 0.98$, $p = .328$, $\beta = -.202$. Thus, there was a positive association between self-enhancement scores and AB only for the non-GD EGM players. The impact of self-enhancement on non-GD EGM players' AB was fairly substantial. Although the overall AB of non-GD EGM players was significantly smaller than the overall AB of GD-EGM players (as described previously), the moderation analysis indicated that the mean AB of non-GD EGM players with high self-enhancement scores (609 ms) was not significantly different than the mean AB of GD-EGM players with high self-enhancement scores (846 ms), $t(75) = 0.61$, $p = .541$. Only when self-enhancement scores were low or average was there a group difference in AB, $t(75) = 4.02$, $p < .001$, and $t(75) = 3.34$, $p = .001$, respectively.

Figure 4 shows the relationship between GACS anticipation subscores and AB. This interaction reflected group differences in the relationship between GACS anticipation scores and AB. For non-GD EGM players, higher anticipation scores were associated with a larger AB, whereas for GD-EGM players and non-gamblers, higher scores had little impact on AB. An analysis of the data of the non-GD and GD-EGM players confirmed that the interaction was present when the analysis was confined to EGM players, $F(1, 73) = 4.888$, $p = .030$ ($\Delta R^2 = 4.95\%$). A simple slopes analysis showed that for non-GD EGM players, higher anticipation scores were associated with a larger AB, $t(51) = 3.30$, $p = .001$, $\beta = .430$, whereas for GD-EGM players this was not the case, $t(24) = 0.55$, $p = .580$, $\beta = -.111$. Thus, only for the non-GD EGM players was there a positive association between anticipation scores and AB. Comparisons between the ABs of non-GD EGM and GD-EGM players at the three levels of anticipation scores (low, average, and high) showed that the two groups differed when anticipation scores were low

Figure 2

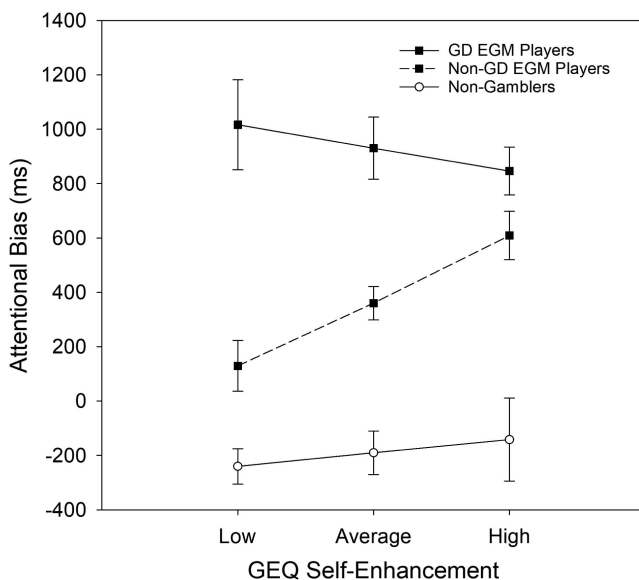
Attentional Bias (in ms) as a Function of GEQ Enjoyment Scores for Non-EGM Players, Non-GD EGM Players, and GD EGM Players



Note. EGM = electronic gaming machines; GD = gambling disorder; GEQ = gambling expectancy questionnaire.

Figure 3

Attentional Bias (in ms) as a Function of GEQ Self-Enhancement Scores for Non-EGM Players, Non-GD EGM Players, and GD EGM Players

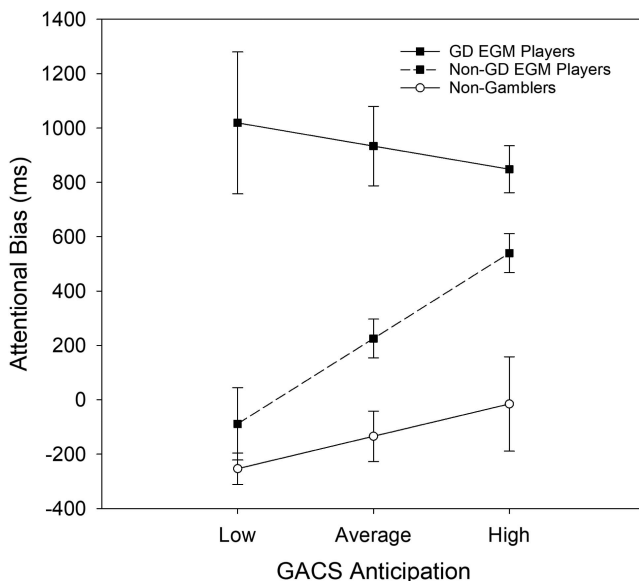


Note. EGM = electronic gaming machines; GD = gambling disorder; GEQ = gambling expectancy questionnaire.

$t(75) = 3.59, p < .001$, and average, $t(75) = 3.43, p = .001$, but not when they were high, $t(75) = 0.92, p = .355$. These comparisons provide additional evidence that anticipation-related cravings had a strong influence on the AB of non-GD EGM players.¹

Figure 4

Attentional Bias (in ms) as a Function of GACS Anticipation Scores for Non-EGM Players, Non-GD EGM Players, and GD EGM Players



Note. EGM = electronic gaming machines; GD = gambling disorder; GACS = gambling craving scale.

Discussion

Preferential attention to addiction-related stimuli (i.e., attentional bias) has been well documented in substance use disorders (Field & Cox, 2008). In contrast, much less is known of ABs for behavioral addictions such as gambling. In the present study, we compared the ABs for EGM images in a sample of non-gamblers, non-GD EGM players, and GD-EGM players. Our attention data were collected using an eye-gaze tracking paradigm, which has been suggested as the gold standard paradigm for the assessment of ABs (Field & Cox, 2008). In addition, we extended the literature on this topic by exploring gambling and psychological correlates and moderators of ABs among both groups of EGM players.

The results of our study provided further support for the existence of an AB in individuals who gamble with the sample in our study consisting of GD and non-GD EGM players. Whereas non-gamblers exhibited an AB for neutral images, both groups of EGM players preferentially attended to EGM images, with GD EGM players having the larger AB. Interestingly, while GD and non-GD EGM players did not differ in their time of first fixation to the EGM images, GD-EGM players had longer total fixation times to these images. Combined, these results indicate that AB in GD is defined by sustained attention to gambling-related stimuli rather than faster attentional capture. Individuals with GD may have difficulty shifting attention away from gambling-related stimuli, which may contribute to gambling-focused cognitions and in a way that is consistent with the incentive sensitization theory.

These findings have several important theoretical and clinical implications. First, the results provide further support for the similarities between substance use and behavioral addictions and therefore increase confidence in the recent classification of GD in the Substance-Related and Addictive Disorders Section of the DSM-5 (American Psychiatric Association, 2013). In terms of theoretical implications, the results support the possibility that the incentive-sensitization theory may be relevant not only to GD but perhaps to other putative behavioral addictions as well (see Robinson & Berridge, 2008).

With respect to psychological correlates of AB, our results revealed significant associations between AB and gambling expectancies (with moderate effect sizes), specifically, the enjoyment/arousal and emotional aspects, as well as coping and financial motives. It is interesting to note that the enjoyment/arousal aspect reflects a positive expectancy of gambling, whereas the emotional impact reflects a negative expectancy. This result could reflect the compulsive nature of gambling, given one of the hallmark characteristics of GD is the continued engagement in gambling despite negative consequences (American Psychiatric Association, 2013). Thus, it is perhaps not surprising that those who gamble regularly and develop an AB may be more aware of the potential negative consequences of gambling given their experiences, while at the same time may also hold positive expectancies related to gambling (e.g., winning).

¹ The correlational and moderation analyses were conducted using the AB variable and not the time of first fixation variable because we did not expect strong associations between the first fixation times and the psychological variables included in our models (given that first fixation times do not reflect sustained attention). Post-hoc moderation analyses were conducted with the time of first fixation data and, as expected, none of the interactions were significant (all $ps > .10$).

With respect to gambling motives, AB was significantly correlated with coping and financial motives. Both coping and financial motives have been associated with increased GD severity (Kim et al., 2019; Tabri et al., 2015). As such, this result could reflect that those who are motivated to gamble for coping and financial motives are more likely to be highly engaged in gambling, thereby making it more likely for operant conditioning processes to culminate in the development of an AB. Lastly, alcohol use severity was significantly associated with AB. This finding could be due to the significant co-morbidity between alcohol and gambling (Dowling et al., 2015) and the development of cross-cue reactivity between the two. Indeed, there is some evidence to suggest cross-cue reactivity may develop between gambling and alcohol (e.g., Zack et al., 2005). Future research examining cross-cue reactivity using an eye-tracking paradigm among people who gamble and who also drink would be highly informative.

We also found several significant gambling-related variables that moderated the relationship between EGM gambling status and AB. Specifically, positive expectancies related to gambling (the enjoyment/arousal and self-enhancement anticipation subscales of the GEQ) and the anticipation subscale of the GACS were significant moderators. On the other hand, enhancement and financial motives for gambling were not significant moderators. Interestingly, the moderation analyses revealed that the associations between AB and the enjoyment/arousal and anticipation were more pronounced for non-GD EGM players compared to GD-EGM players. That is, the moderating effect on AB was greater for EGM players who reported less problematic gambling. This outcome suggests that positive expectancies are important factors in AB at the early stages of GD but may not influence AB when GD has fully developed. In other words, AB may be relatively automatic in GD, possibly due to the shift in hedonistic “liking” during the early stages of problem gambling and then to compulsive “wanting” when GD develops. From an incentive-sensitization perspective, these findings make intuitive sense. For participants who have not developed GD, positive expectancies will be associated with AB as there are hedonistic pleasures associated with gambling, whereas for those with GD, gambling is no longer associated with liking and therefore, positive expectancies are not associated with AB. It will be important to replicate these findings to determine if they provide support for the application of the incentive-sensitization theory to GD.

Of course, there are other explanations for the associations between positive gambling expectancies and AB. For example, it is possible that people with high positive expectancies related to gambling are at risk of developing GD due to high levels of attentional bias. If so, further studies will be necessary to assess this potential risk factor of GD. Another possibility is that non-GD EGM players may not have experienced negative consequences as a result of their EGM play, and thus may be more attentive to EGM-related cues, which may be associated with future EGM play. Conversely, GD-EGM players are likely to have experienced negative consequences as a result of their gambling, and despite holding positive expectancies toward gambling, they may actively suppress their attention to EGM-related stimuli to avoid increased cravings and urges to gamble. Our results identify several important research questions for future investigations of the psychological predictors and moderators of AB in GD.

The existence of ABs in EGM gamblers, and associations with key psychological variables, provides a strong rationale for the

continued development of attentional bias modification programs for gambling, similar to the paradigms that have been developed for substance use disorders (e.g., Attwood et al., 2008; Schoenmakers et al., 2010). In an attentional bias modification training paradigm, participants are shown pairs of images (e.g., an addiction-related image paired with a neutral image), and the task contingencies are designed to encourage participants to habitually shift their attention away from the addiction-related cues. Reducing the gambling-related AB in this manner is thought to assist in the reduction of cravings and urges. Although there is some empirical support for the use of attentional bias modification as an intervention for substance use disorders (Field et al., 2014; Heitmann et al., 2018), much more research is required to establish its efficacy, and specifically for individuals with GD.

Limitations

The limitations of this study point to several directions for future research. Our sample consisted of community and university students who were not seeking treatment for gambling, and therefore our results may not generalize to those seeking such treatment. That said, our results may have wider generalizability, given that less than 10% of people with GD seek treatment (Hodgins et al., 2011). On the other hand, our sample consisted of mainly white community-dwelling individuals and may not generalize to non-white individuals. Our study used a cross-sectional design, so cause-and-effect relationships between the psychosocial variables and ABs cannot be unequivocally established. It is also possible that temporal associations between AB and specific psychological factors may differ. For example, positive expectancies related to gambling may lead to an AB, whereas an AB may lead to negative expectancies related to gambling. Another limitation is that our sample of gamblers was restricted to EGM players. We restricted our sample to one type of gambler given recent findings suggesting the specificity of ABs based on the preferred type of gambling (McGrath et al., 2018). Thus, the extent to which the present results generalize to those whose preferred form of gambling is not EGMs (e.g., poker players) will need to be assessed in future studies that use different samples of GD. With respect to the stimuli we used, some of the neutral images may be associated with positive valence (e.g., the arcade game terminals), which may have attenuated the AB for EGM images and led us to underestimate the magnitude of this bias. Researchers should carefully weigh such considerations when selecting neutral or control images in future studies. The relatively small sample of individuals with GD ($n = 25$) may have reduced our ability to detect modest correlations and interaction effects and researchers should keep this in mind when interpreting our findings. Additional studies with larger, more diverse samples will increase confidence in our findings and conclusions.

Conclusion

ABs may play an important role in the etiology and maintenance of addictive disorders (Field & Cox, 2008). In the present study, we provide further evidence that an AB may also develop in the context of behavioral addictions, and specifically gambling. The results of our study expand our understanding of the etiology and maintenance of gambling and may contribute to improving treatment options for the significant number of individuals who develop GD.

References

- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders (DSM-5®)*.
- Attwood, A. S., O'Sullivan, H., Leonards, U., Mackintosh, B., & Munafò, M. R. (2008). Attentional bias training and cue reactivity in cigarette smokers. *Addiction*, 103(11), 1875–1882. <https://doi.org/10.1111/j.1360-0443.2008.02335.x>
- Beck, A. T., Steer, R. A., & Brown, G. K. (1996). *Manual for the Beck Depression Inventory–II*. Psychological Corporation.
- Berridge, K. C., & Robinson, T. E. (2016). Liking, wanting, and the incentive-sensitization theory of addiction. *American Psychologist*, 71(8), 670–679. <https://doi.org/10.1037/amp0000059>
- Ciccarelli, M., Cosenza, M., Griffiths, M. D., Nigro, G., & D'Olimpio, F. (2019). Facilitated attention for gambling cues in adolescent problem gamblers: An experimental study. *Journal of Affective Disorders*, 252, 39–46. <https://doi.org/10.1016/j.jad.2019.04.012>
- Ciccarelli, M., Nigro, G., Griffiths, M. D., Cosenza, M., & D'Olimpio, F. (2016a). Attentional biases in problem and non-problem gamblers. *Journal of Affective Disorders*, 198, 135–141. <https://doi.org/10.1016/j.jad.2016.03.009>
- Ciccarelli, M., Nigro, G., Griffiths, M. D., Cosenza, M., & D'Olimpio, F. (2016b). Attentional bias in non-problem gamblers, problem gamblers, and abstinent pathological gamblers: An experimental study. *Journal of Affective Disorders*, 206, 9–16. <https://doi.org/10.1016/j.jad.2016.07.017>
- Cohen, J. (1969). *Statistical power analysis for the behavioral sciences*. Academic Press.
- Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112, 155–159. <https://doi.org/10.1037/0033-2909.112.1.155>
- Copersino, M. L., Serper, M. R., Vadhan, N., Goldberg, B. R., Richarme, D., Chou, J. C. Y., Stitzer, M., & Cancro, R. (2004). Cocaine craving and attentional bias in cocaine dependent schizophrenic patients. *Psychiatry Research*, 128, 209–218. <https://doi.org/10.1016/j.psychres.2004.07.006>
- Crombez, G., Van Ryckeghem, D. M., Eccleston, C., & Van Damme, S. (2013). Attentional bias to pain-related information: A meta-analysis. *Pain*, 154(4), 497–510. <https://doi.org/10.1016/j.pain.2012.11.013>
- Currie, S. R., Hodgins, D. C., & Casey, D. M. (2013). Validity of the problem gambling severity index interpretive categories. *Journal of Gambling Studies*, 29(2), 311–327. <https://doi.org/10.1007/s10899-012-9300-6>
- Dechant, K. (2014). Show me the money: Incorporating financial motives into the Gambling Motives Questionnaire. *Journal of Gambling Studies*, 30(4), 949–965. <https://doi.org/10.1007/s10899-013-9386-5>
- Delfabbro, P., King, D. L., Browne, M., & Dowling, N. A. (2020). Do EGMs have a stronger association with problem gambling than racing and casino table games? Evidence from a decade of Australian prevalence studies. *Journal of Gambling Studies*, 36, 499–511. <https://doi.org/10.1007/s10899-020-09950-5>
- Dowling, N. A., Cowlshaw, S., Jackson, A. C., Merkouris, S. S., Francis, K. L., & Christensen, D. R. (2015). Prevalence of psychiatric co-morbidity in treatment-seeking problem gamblers: A systematic review and meta-analysis. *Australian and New Zealand Journal of Psychiatry*, 49(6), 519–539. <https://doi.org/10.1177/0004867415575774>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39, 175–191. <https://doi.org/10.3758/BF03193146>
- Ferris, J., & Wynne, H. (2001). *The Canadian problem gambling index: Final report*. Submitted for the Canadian Centre on Substance Abuse.
- Field, M. (2005). Cannabis 'dependence' and attentional bias for cannabis-related words. *Behavioural Pharmacology*, 16, 473–476. <https://doi.org/10.1097/00008877-200509000-00021>
- Field, M., & Cox, W. M. (2008). Attentional bias in addictive behaviors: A review of its development, causes, and consequences. *Drug and Alcohol Dependence*, 97(1–2), 1–20. <https://doi.org/10.1016/j.drugalcdep.2008.03.030>
- Field, M., Hogarth, L., Bleasdale, D., Wright, P., Fernie, G., & Christiansen, P. (2011). Alcohol expectancy moderates attentional bias for alcohol cues in light drinkers. *Addiction*, 106(6), 1097–1103. <https://doi.org/10.1111/j.1360-0443.2011.03412.x>
- Field, M., Marhe, R., & Franken, I. H. (2014). The clinical relevance of attentional bias in substance use disorders. *CNS Spectrums*, 19(3), 225–230. <https://doi.org/10.1017/S1092852913000321>
- Field, M., Munafò, M. R., & Franken, I. H. (2009). A meta-analytic investigation of the relationship between attentional bias and subjective craving in substance abuse. *Psychological Bulletin*, 135(4), 589–607. <https://doi.org/10.1037/a0015843>
- Gillespie, M., Derevensky, J., & Gupta, R. (2007). The utility of outcome expectancies in the prediction of adolescent gambling behavior. *Journal of Gambling Issues*, 19, 69–85. <https://doi.org/10.4309/jgi.2007.19.4>
- Grant, L. D., & Bowling, A. C. (2015). Gambling attitudes and beliefs predict attentional bias in non-problem gamblers. *Journal of Gambling Studies*, 31(4), 1487–1503. <https://doi.org/10.1007/s10899-014-9468-z>
- Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis* (2nd ed.). The Guilford Press.
- Hayes, A. F., & Rockwood, N. J. (2017). Regression-based statistical mediation and moderation analysis in clinical research: Observations, recommendations, and implementation. *Behaviour Research and Therapy*, 98, 39–57. <https://doi.org/10.1016/j.brat.2016.11.001>
- Heitmann, J., Bennik, E. C., van Hemel-Ruiter, M. E., & de Jong, P. J. (2018). The effectiveness of attentional bias modification for substance use disorder symptoms in adults: A systematic review. *Systematic Reviews*, 7(1), 160. <https://doi.org/10.1186/s13643-018-0822-6>
- Hodgins, D. C., Stea, J. N., & Grant, J. E. (2011). Gambling disorders. *The Lancet*, 378(9806), 1874–1884. [https://doi.org/10.1016/S0140-6736\(10\)62185-X](https://doi.org/10.1016/S0140-6736(10)62185-X)
- Hønsi, A., Mentzoni, R. A., Molde, H., & Pallesen, S. (2013). Attentional bias in problem gambling: A systematic review. *Journal of Gambling Studies*, 29(3), 359–375. <https://doi.org/10.1007/s10899-012-9315-z>
- Hudson, A., Olatunji, B. O., Gough, K., Yi, S., & Stewart, S. H. (2017). Eye on the prize: High-risk gamblers show sustained selective attention to gambling cues. *Journal of Gambling Issues*, 34, 100–119. <https://doi.org/10.4309/jgi.2016.34.6>
- IBM Corp. (2019). *IBM SPSS Statistics for Windows, Version 26.0*. (Released 2019)
- Jones, A., Hogarth, L., Christiansen, P., Rose, A. K., Martinovic, J., & Field, M. (2012). Reward expectancy promotes generalized increases in attentional bias for rewarding stimuli. *The Quarterly Journal of Experimental Psychology*, 65(12), 2333–2342. <https://doi.org/10.1080/17470218.2012.686513>
- Kim, H. S., Poole, J. C., Hodgins, D. C., McGrath, D. S., & Dobson, K. S. (2019). Betting to deal: Coping motives mediate the relationship between urgency and Problem gambling severity. *Addiction Research & Theory*, 27(2), 95–103. <https://doi.org/10.1080/16066359.2018.1455188>
- Lorains, F. K., Cowlshaw, S., & Thomas, S. A. (2011). Prevalence of comorbid disorders in problem and pathological gambling: Systematic review and meta-analysis of population surveys. *Addiction*, 106(3), 490–498. <https://doi.org/10.1111/j.1360-0443.2010.03300.x>
- Mann, K., Fauth-Bühler, M., Higuchi, S., Potenza, M. N., & Saunders, J. B. (2016). Pathological gambling: A behavioral addiction. *World Psychiatry*, 15(3), 297–298. <https://doi.org/10.1002/wps.20373>
- Marissen, M. A., Franken, I. H., Waters, A. J., Blanken, P., Van Den Brink, W., & Hendriks, V. M. (2006). Attentional bias predicts heroin relapse following treatment. *Addiction*, 101(9), 1306–1312. <https://doi.org/10.1111/j.1360-0443.2006.01498.x>
- Martens, M. P., Rocha, T. L., Cimini, M. D., Diaz-Myers, A., Rivero, E. M., & Wulfert, E. (2009). The co-occurrence of alcohol use and gambling activities in first-year college students. *Journal of American College Health*, 57(6), 597–602.

- McGrath, D. S., Barrett, S. P., Stewart, S. H., & Schmid, E. A. (2012). The effects of acute doses of nicotine on video lottery terminal gambling in daily smokers. *Psychopharmacology*, 220(1), 155–161. <https://doi.org/10.1007/s00213-011-2465-3>
- McGrath, D. S., Meitner, A., & Sears, C. R. (2018). The specificity of attentional biases by type of gambling: An eye-tracking study. *PLOS ONE*, 13(1), Article e0190614. <https://doi.org/10.1371/journal.pone.0190614>
- McGrath, D. S., Sears, C. R., Fernandez, A., & Dobson, K. S. (2020). Attentional biases in low-risk and high-risk gamblers and the moderating effect of daily psychosocial stress. *Addiction Research & Theory*. <https://doi.org/10.1080/16066359.2020.1762867>
- Mogg, K., & Bradley, B. P. (2002). Selective processing of smoking-related cues in smokers: Manipulation of deprivation level and comparison of three measures of processing bias. *Journal of Psychopharmacology*, 16, 385–392. <https://doi.org/10.1177/026988110201600416>
- Reinert, D. F., & Allen, J. P. (2007). The alcohol use disorders identification test: An update of research findings. *Alcoholism: Clinical and Experimental Research*, 31(2), 185–199. <https://doi.org/10.1111/j.1530-0277.2006.00295.x>
- Robinson, T. E., & Berridge, K. C. (2008). The incentive sensitization theory of addiction: Some current issues. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 363(1507), 3137–3146. <https://doi.org/10.1098/rstb.2008.0093>
- Saunders, J. B., Aasland, O. G., Babor, T. F., De la Fuente, J. R., & Grant, M. (1993). Development of the alcohol use disorders identification test (AUDIT): WHO collaborative project on early detection of persons with harmful alcohol consumption-II. *Addiction*, 88(6), 791–804. <https://doi.org/10.1111/j.1360-0443.1993.tb02093.x>
- Schellenberg, B. J., McGrath, D. S., & Dechant, K. (2016). The Gambling Motives Questionnaire financial: Factor structure, measurement invariance, and relationships with gambling behaviour. *International Gambling Studies*, 16(1), 1–16. <https://doi.org/10.1080/14459795.2015.1088559>
- Schluter, M. G., Kim, H. S., Poole, J. C., Hodgins, D. C., McGrath, D. S., Dobson, K. S., & Taveres, H. (2019). Gambling-related cognitive distortions mediate the relationship between depression and disordered gambling severity. *Addictive Behaviors*, 90, 318–323. <https://doi.org/10.1016/j.addbeh.2018.11.038>
- Schoenmakers, T. M., de Bruin, M., Lux, I. F., Goertz, A. G., Van Kerkhof, D. H., & Wiers, R. W. (2010). Clinical effectiveness of attentional bias modification training in abstinent alcoholic patients. *Drug and Alcohol Dependence*, 109(1–3), 30–36. <https://doi.org/10.1016/j.drugalcdep.2009.11.022>
- Sears, C., Quigley, L., Fernandez, A., Newman, K., & Dobson, K. (2019). The reliability of attentional biases for emotional images measured using a free-viewing eye-tracking paradigm. *Behavior Research Methods*, 51(6), 2748–2760. <https://doi.org/10.3758/s13428-018-1147-z>
- Sharma, D., Albery, I. P., & Cook, C. (2001). Selective attentional bias to alcohol related stimuli in problem drinkers and non-problem drinkers. *Addiction*, 96, 285–295. <https://doi.org/10.1046/j.1360-0443.2001.96228512.x>
- Stewart, S. H., & Zack, M. (2008). Development and psychometric evaluation of a three-dimensional Gambling Motives Questionnaire. *Addiction*, 103(7), 1110–1117. <https://doi.org/10.1111/j.1360-0443.2008.02235.x>
- Tabri, N., Dupuis, D. R., Kim, H. S., & Wohl, M. J. (2015). Economic mobility moderates the effect of relative deprivation on financial gambling motives and disordered gambling. *International Gambling Studies*, 15(2), 309–323. <https://doi.org/10.1080/14459795.2015.1046468>
- Thomsen, K. R., Fjorback, L. O., Møller, A., & Lou, H. C. (2014). Applying incentive sensitization models to behavioral addiction. *Neuroscience and Biobehavioral Reviews*, 45, 343–349. <https://doi.org/10.1016/j.neubiorev.2014.07.009>
- Weinstock, J., Whelan, J. P., & Meyers, A. W. (2004). Behavioral assessment of gambling: An application of the timeline followback method. *Psychological Assessment*, 16, 72–80. <https://doi.org/10.1037/1040-3590.16.1.72>
- Young, M. M., & Wohl, M. J. (2009). The Gambling Craving Scale: Psychometric validation and behavioral outcomes. *Psychology of Addictive Behaviors*, 23(3), 512–522. <https://doi.org/10.1037/a0015043>
- Zack, M., Stewart, S. H., Klein, R. M., Loba, P., & Fragopoulos, F. (2005). Contingent gambling-drinking patterns and problem drinking severity moderate implicit gambling-alcohol associations in problem gamblers. *Journal of Gambling Studies*, 21(3), 325–354. <https://doi.org/10.1007/s10899-005-3102-z>

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